

Building AI Literacy: Adolescents’ long-term interactions with social robots enhance AI literacy and motivation

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Figure 1. A) Schematic overview of Marty robot build guide. B-D) Deployment pictures of students building the Marty robot. E-G) Deployment pictures of students interacting with Marty.

ABSTRACT

Social robots hold significant promise for classroom education across a variety of subjects, including STEM, reading, and AI literacy. However, high costs, technical complexity, and a lack of long-term engagement hinder widespread adoption. While low-cost “do-it-yourself” (DIY) robotic toolkits offer a possible solution, it remains unclear how physical assembly impacts students’ learning outcomes, motivation, and perceptions of the robot. In this preregistered, seven-week longitudinal study, 61 adolescent students aged 12–16 years old either assembled the humanoid social robot Marty before interacting with it or interacted with a pre-assembled Marty robot within an AI literacy curriculum. We measured students’ AI literacy, intrinsic motivation, and perceptions of socialness across four time points. Both interaction conditions led to short- and long-term AI literacy gains, with no significant differences between groups. However, students in the building condition reported higher intrinsic motivation throughout the study and rated it as more useful and enjoyable. Contrary to the ‘IKEA effect’, in which adults and younger children frequently evaluate self-built artifacts more positively, our results provide no evidence that assembly alters adolescents’ perceptions of robot socialness. Our findings indicate that DIY robotic toolkits may support student engagement and learning without compromising a robot’s social effectiveness, offering a scalable and motivationally advantageous pathway for educational robot deployment.

CCS CONCEPTS

Human-centered computing→Empirical studies in HCI; •

Social and professional topics→K-12 education; Computing literacy

KEYWORDS

Social robots, AI literacy, K-12 education, Hands-on learning, Perception

ACM Reference format:

FirstName Surname, FirstName Surname and FirstName Surname. 2018. Insert Your Title Here: Insert Subtitle Here. In *Proceedings of ACM Woodstock conference (WOODSTOCK’18)*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/1234567890>

1 Introduction and related work

The adoption of (generative) Artificial Intelligence (AI) has reached up to 84% amongst adolescents and young adults, indicating that AI adoption has accelerated and is rapidly becoming an integral part of young people’s lives [57, 88]. However, children often lack the critical skills required to interact with AI safely, responsibly, and effectively [32, 39, 97, 117]. Because mere exposure is not sufficient to foster AI literacy [61, 76], an urgent need exists for evidence-based learning interventions that guide adolescents from passive consumers to critical consumers and creators of AI technology.

To address this challenge, we conducted one of the first long-term educational HRI studies with adolescents (aged 12-16 years) examining how building and interacting versus only interacting with a pre-built social robot affects learning, motivation, and robot perceptions. We employed an active control group design with a *build group*, who assembled their own “do-it-yourself” (DIY) humanoid Marty robots [132], and a *control*

group, who interacted with already assembled Marty robots during a seven-week AI literacy curriculum. We aimed to assess the effect of robot-building on: students' objective and subjective AI literacy learning (RQ1); intrinsic motivation (RQ2); robot perception (RQ3); and attachment and intention to use the robot (RQ4). We additionally assessed the practicalities of implementing such a program in the classroom (RQ5). Based on the reviewed literature, we hypothesized greater learning and more positive perception outcomes for students who built their robot before interacting with it. All research questions and hypotheses were preregistered and can be found in **Supp. A** and the OSF project [BLINDED]. This paper offers several contributions for HRI researchers and educational practitioners:

- **Empirical & Pedagogical Viability:** Provides a 7-week longitudinal evaluation demonstrating that adolescents' HRI yields long-term AI literacy gains, with a motivational benefit and higher intention to use for those in the robot-assembly over robot-interaction-only group.
- **Reframing the 'IKEA Effect' & Social Perception:** Challenges the universality of the IKEA effect (i.e., assembly leading to more positive evaluations) in HRI by finding no evidence for improved robot evaluations through assembly in adolescents, and instead identifying perceived anthropomorphism as the primary driver of social ratings.
- **Scalable Classroom Deployment:** Proposes low-cost DIY robotic toolkits as innovative, accessible, and effective alternatives to expensive, complex robots based on mixed-methods student and teacher feedback.

2 Related work

2.1 Hands-on AI literacy and DIY social robotics

Rooted in the educational concepts of constructivism and constructionism [1, 85, 87], hands-on learning allows learners to explore technical systems through physical manipulation, iterative building, and reflection [18, 58, 109]. Social robots offer a particularly promising means for hands-on AI literacy education because they integrate physical hardware, programmable software, and social interaction within a single system [112]. Previous research has established that social robots can support children's cognitive and affective learning across different domains and roles [10, 13, 127]. In such contexts, social robots have been deployed as tutors, peers, or teachers to support learning across domains including mathematics [65, 95], language learning [35, 51, 118], reading [19, 22, 67, 78], handwriting [113, 120], and, more recently, AI literacy [119, 125].

However, the common practice of deploying social robots in roles such as tutors or peers [13, 127] positions children as users of finished robotic and AI systems rather than as creators who engage with how these systems are built, programmed, and constrained. Consequently, robotic learning interventions often present children with black-box systems and ask them to use the technology as a means to learn about AI, without treating the technology itself as a learning opportunity. Emerging DIY robotic toolkits, such as the Reachy mini [92] or Blossom [107] robots, offer accessible alternatives and align with recent calls to broaden the roles of social robots [91, 93, 128]. These platforms allow

students to actively build, wire, calibrate, and program social agents, turning users into active creators of embodied AI at a fraction ($\sim <3\%$) of the cost of the widely available pre-assembled commercial systems that are often the focus of research in this domain [e.g., NAO, Pepper, 8, 66].

2.2 The Demystification Trade-Off: Authorship vs. Social Perception

DIY toolkits offer several advantages, but transitioning students from mere consumers to creators of embodied AI technology may introduce a psychological trade-off. On the one hand, the process of assembly is well-documented to improve users' evaluation of objects. Labor and self-assembly frequently enhance product evaluation via the 'IKEA effect' [82, 86] and the 'I designed it myself' effect [36, 108]. In adults' interactions with robots, numerous studies have demonstrated that the sense of ownership and feelings of accomplishment derived from assembling a robot positively impact perceived friendliness, team cohesion, emotional attachment, empathy, and user self-efficacy [40, 43, 108, 114], which may in turn support sustained engagement with a robot [131]. On the other hand, requiring students to assemble a robot forces them to see behind the curtain. Exposing motors, wires, and mechanical constraints risks demystifying the system, potentially shattering the illusion of social agency, lowering social attributions, and diminishing pedagogical effectiveness [39]. Beyond mechanical demystification, assembly shifts students from consumers to creators, which may attenuate perceptions of intelligence. While people often expect AI tools to be smarter or more knowledgeable than themselves [63], builders may implicitly assume that a system cannot be smarter than the person who built it.

2.3 Robot assembly and perception in adolescents

Despite the potential tension between psychological ownership and socialness perception, very few studies have investigated the effect of assembly on *school-aged children's* perceptions of robots. Existing studies have primarily focused on preschool-aged children (4–6 years), demonstrating that short-term robot assembly and AI tasks can increase perceived robot intelligence and socialness, particularly among children with higher baseline AI knowledge [106, 126]. However, findings from these studies remain preliminary, as the studies featured small samples and did not conduct formal statistical testing. Additionally, findings from early childhood cannot be directly generalized to *adolescents* (aged 12–16 years) for whom the risks and opportunities of AI use are arguably even greater. Adolescents exhibit distinct developmental trajectories in robot perception [60, 68, 79], and the IKEA effect may operate differently in children and adults [71]. Given that adolescents are the largest user group of (generative) AI technologies but remain especially vulnerable to safety and privacy risks [97, 117], investigating how hands-on robot assembly affects their AI literacy and social robot perception is essential for designing scalable, safe, and effective educational HRI interventions.

3 Methods and materials

3.1 Participants

We collected data from 61 students (12-16 years; $M_{age} = 13.40$, $SD = 0.75$; 15 girls, 42 boys, 4 other) and two female teachers ($M_{age} = 45.50$ years, $SD = 4.95$) across two classes at a rural German comprehensive school spanning low, medium, and high academic tracks. Informed teacher/parental consent, student assent (≥ 14 years), and photo permissions were collected. Study procedures were approved by the local ethics board (blinded).

3.2 Design and procedure

We conducted a seven-week, between-subjects, active-control study (2x4 design: condition (Building + interaction vs. Interaction-only) x timepoint (T1, T4, T7, T8)) during Natural Sciences and Informatics classes (**Figure 2**). Students worked in randomized, fixed teams of 3-4, aligning with Chi's collaborative learning taxonomy [23]. In Phase 1 (weeks 1-4), the Build group ($n = 31$) assembled the Marty robot [132] over 4 hours using a build guide [17]. The active-control Interaction-only group ($n = 30$) played non-programming controller-based games (e.g., soccer, obstacle courses, escape games) with pre-assembled Marty robots to ensure equal interaction time without providing an AI knowledge advantage (**Figure 2**). In Phase 2 (weeks 5-7), both groups completed identical learning activities with their robots.

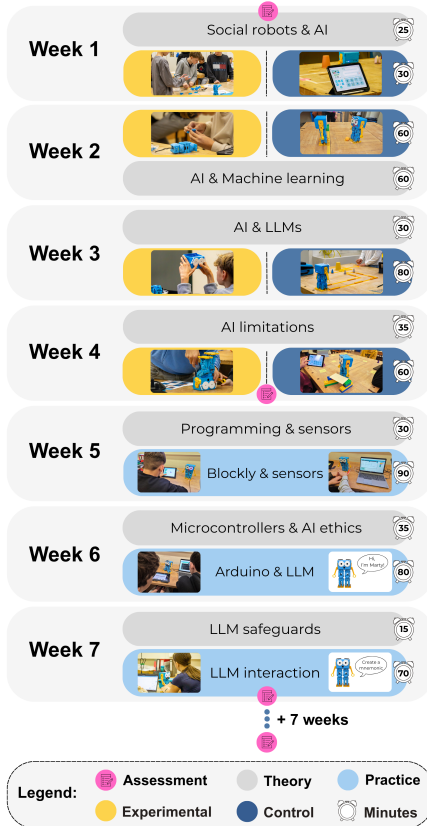


Figure 2. Schematic overview of the seven-week curriculum.

Two teachers instructed two parallel classrooms ($n \sim 16$) in both the Build condition (Nov–Dec 2025) and Interaction condition (Jan–Mar 2026) due to the school's classroom set-up with two

adjacent rooms. Surveys (~15 minutes) were administered at pre-week (week 1), mid- (week 4), post- (week 7), and retention test (7-week follow-up at week 14) via Qualtrics [133]. Surveys began with watching a 5-second Marty video, followed by counterbalanced measures of perception [RoSAS-SF, 21, SOAS, 27, intentional acceptance scale, 30, product-consumer attachment scale, 98], subjective AI literacy [MAILS; 50], and objective AI literacy (23 self-developed items). Brief weekly surveys (~10 minutes) evaluated students' intrinsic motivation [Intrinsic Motivation Scale, 41], and lesson feedback from students and teachers (**Supp. H**). Students were debriefed post-study; teachers received a €100 classroom voucher. Study design, hypotheses, and analysis plans were preregistered, and all data/materials are available on OSF [BLINDED].

3.3 Curriculum

The seven-week curriculum was co-developed with teachers, based on prominent AI literacy curricula [2-7, 25, 29, 33, 77, 83, 94, 110, 111], and covered social robotics, Artificial Intelligence (AI), sensors, microcontrollers, Large Language Models (LLMs), and programming (**Figure 2**). Each weekly session included theoretical content and practical activities and lasted 140 minutes. The total experimental time in the classroom was 16 hours and 33 minutes. The curriculum is available on GitHub [BLINDED], and detailed information regarding each session's content and rationale is available in **Supp. B**.

3.4 Marty robot

Marty is a ~25 cm desktop humanoid robot designed for science education (~400€) [132]. Equipped with two LED eyes, 11 motors, and five sensors, Marty is typically sold pre-assembled but can also be shipped as parts and built using a guide (**Figure 1**, 17). Assembly includes wiring components, assembling plastic parts, mounting limbs, and calibrating motors. Once built, Marty supports diverse behaviors (e.g., walking, dancing, speaking) programmed via screen-free, block-based, or text-based coding (Python/C++) using an app or web interface.

3.5 Measures

More information on scale selection, adaptations, reliability, and complete item sets are available in **Supp. C** and on OSF [BLINDED].

3.5.1 Learning (RQ1 & RQ2)

3.5.1.1 Subjective AI literacy. We assessed self-reported AI literacy using an age-reviewed, 10-item short version of the validated Meta AI literacy scale [MAILS; 20, 50]. Items (e.g., "I can tell if I am dealing with an application based on artificial intelligence") were rated on an 11-point scale ranging from 0 (*not at all*) to 10 (*very well*). The reliability of the scale and its subscales has been established [$\alpha > .90$, 50].

3.5.1.2 Objective AI literacy. To complement subjective self-reports, which may not accurately reflect students' actual capabilities [24, 70, 122], we assessed objective AI literacy. Objective knowledge was evaluated using 23 self-developed multiple-choice items spanning three curriculum domains: general AI knowledge (13 items), technical literacy (6 items), and programming literacy (4 items). Each item provided four options plus an "I don't know" option, resulting in total scores from 0 to 23. Two attention checks were embedded within the survey.

Across the three subscales, item difficulty ranged from .09 to .58, and "I don't know" selection rates spanned .19 to .67, indicating high overall test difficulty. Corrected item-total correlations demonstrated acceptable discrimination across all items (.11 to .42). Internal consistency was acceptable for general AI Literacy ($\alpha = .65$), but lower for technical ($\alpha = .42$) and programming literacy ($\alpha = .34$).

3.5.1.3 Intrinsic motivation. We assessed students' motivational experiences after each lesson using an 8-item questionnaire adapted from Intrinsic Motivation Inventory frameworks [31, 41, 124]. The adapted scale evaluated five domains on a 5-point scale ranging from 1 (*does not apply*) to 5 (*applies*): enjoyment/interest (two items), perceived competence (two items), pressure/tension (one item), value/usefulness (two items), and information sufficiency/self-appraisal (one item). Cronbach's α was good for the overall *intrinsic motivation* scale ($\alpha = .83$, 95% CI [.80, .85]), the *interest/enjoyment* scale ($\alpha = .83$, 95% CI [.79, .86]), and the *value/usefulness* scale ($\alpha = .83$, 95% CI [.80, .86]), and acceptable for the *perceived competence* scale ($\alpha = .73$, 95% CI [.67, .78]).

3.5.2 Perceptions (RQ3 & RQ4)

3.5.2.1 Robotic social attributes scale (RoSAS-SF). Social perception of Marty was measured using the 6-item validated short RoSAS scale [RoSAS-SF; 21, 54, 80], covering warmth (compassionate, social), competence (competent, reliable), and discomfort (scary, awkward). We additionally added an item for safety, as robot safety is rarely addressed in educational HRI [127] but is especially relevant in interactions with students. Students rated the robot on all seven attributes using a 7-point Likert scale (e.g., "*Marty is compassionate*"; 1 = *not at all*, 7 = *very much so*). The reliability of the RoSAS-SF has been established [warmth: $\alpha = .85$; competence: $\alpha = .92$; discomfort: $\alpha = .68$, 80].

3.5.2.2 Specific object anthropomorphism (SOAS). Perceived anthropomorphism was assessed using the 6-item SOAS scale [e.g., "I feel that this object has likes and dislikes", 27] plus one added item measuring perceived intentionality on a 3-point Likert scale (0 = *no*, 1 = *maybe*, 2 = *yes*). Test-retest and internal reliability of the SOAS have been established [$\omega = .94$, ICC = .78, 27].

3.5.2.3 Consumer-product attachment scale. Attachment to Marty was measured with the validated 5-item consumer-product attachment scale [98]. The scale comprises five items (e.g., "I feel emotionally connected to Marty"), measured on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). The reliability of the scale is excellent [$\alpha = .92$, 49].

3.5.2.4 Intentional acceptance scale. Students' intention to use Marty was measured with the 4-item intentional acceptance scale [30], adapted to a 5-point Likert scale, to match the aforementioned attachment scale. The scale's reliability was found to be acceptable [$\alpha = .72$, 30].

3.5.3 Student and teacher classroom feedback (RQ5). To contextualize classroom feasibility and user experience, we collected student and teacher post-lesson and post-project evaluations using researcher-developed instruments. Students

completed three 5-point Likert items assessing overall project enjoyment, perceived learning, and task difficulty, alongside two open-ended questions regarding liked aspects and suggested changes. Teachers completed an 8-item questionnaire, consisting of five 5-point Likert items (perceived student engagement, relevant question-asking, instructional ease, behavioral management ease, and student learning) and three open-ended prompts (liked aspects, implementation challenges, and general feedback).

3.6 Data analysis

3.6.1 Quantitative data. Outcome variables were aggregated by averaging item ratings (subjective AI literacy, intrinsic motivation, RoSAS subscales, competence, attachment, intention to use) or summing (objective AI literacy, anthropomorphism). To examine changes over time and differences between conditions, we used linear mixed-effects models [LMER; lme4, 12] with nested random effects for participants within triads. We preregistered models with the main effects of condition and timepoint, as well as their interaction. We additionally preregistered exploratory analyses in which we extended the models to include prior knowledge, perceived anthropomorphism, and attachment as covariates. To prevent overfitting and reduce the risk of false positives, we incrementally added effects and ran model comparisons using likelihood ratio tests. We report the results of the best-fit models. *P*-values for fixed effects were computed using Satterthwaite's approximation for degrees of freedom [56]. Where a significant increase in explained variance was observed, we computed post hoc contrasts using estimated marginal means [62] with false discovery rate (FDR) correction. These were evaluated using the same model comparison approach described above. All analysis outputs and any deviations from preregistration can be found in **Supp. D** and on OSF [BLINDED].

3.6.2 Qualitative data. Qualitative feedback was analyzed with an inductive qualitative content analysis, a systematic approach for coding text data into meaningful categories [52, 75], using MaxQDA [74] and Python [89]. Content analysis is a subjective process and is affected by the analysts' experiences and perspectives. To introduce multiple perspectives, we adopted a collaborative approach in which the first author conducted the primary analysis in tandem with a student. Our interpretations were shaped by our team's diverse backgrounds in psychology, business engineering, neuroscience, cognitive sciences, and informatics, bringing together 3 to 20 years of research experience. More information regarding the analysis and coded segments can be found in **Supp. H**.

4 Results

Descriptive statistics and correlations between study measures are reported in **Supp. E**.

4.1 Learning (RQ1 & RQ2)

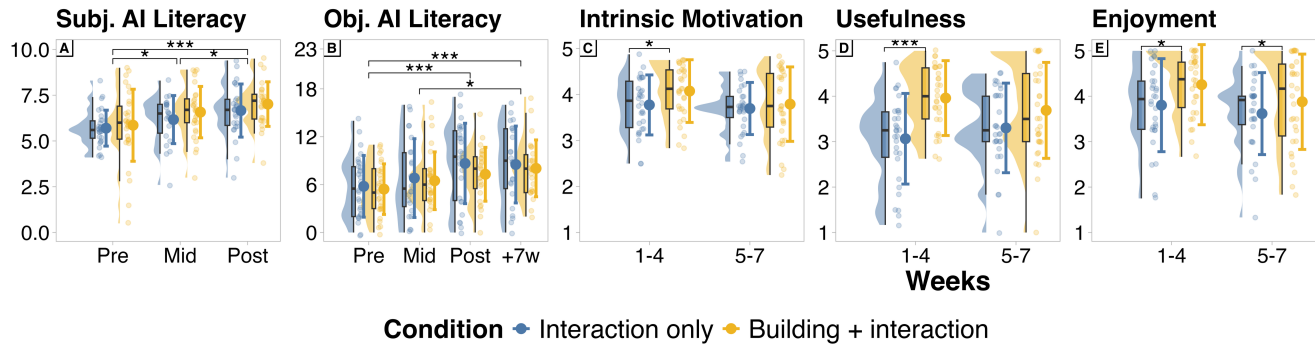


Figure 3. Changes in students' (A) subjective AI literacy, (B) objective AI literacy, (C) intrinsic motivation, (D) interest/enjoyment, and (E) perceived value/usefulness over time and by group. Each plot includes raw data points, means, and standard deviations.

4.1.1 Subjective AI literacy. Likelihood ratio tests showed no evidence that condition ($\chi^2(1) = 0.85, p = .36$) or its interaction with time point ($\chi^2(2) = 0.49, p = .78$) predicted students' AI literacy (**Figure 3A**), indicating no evidence for different learning gains between conditions. However, including timepoint significantly improved model fit ($\chi^2(2) = 23.02, p < .001$): subjective AI literacy scores significantly increased from pre- to mid- ($\beta = 0.56, SE = 0.22, p = .02$) and post-test ($\beta = 1.12, SE = 0.22, p < .001$). Post-hoc comparisons further showed a significant increase in subjective AI literacy scores from mid- to post-test ($\beta = -0.56, SE = 0.23, p = .02$).

4.1.2 Objective AI literacy. Similar to subjective AI literacy, likelihood ratio tests showed no evidence that condition ($\chi^2(1) = 0.14, p = .70$) or its interaction with time point ($\chi^2(3) = 1.07, p = .78$) predicted students' objective AI literacy (**Figure 3B**), providing no evidence for differences in learning between conditions. Including time point significantly improved model fit ($\chi^2(3) = 28.105, p < .001$), indicating that scores significantly increased from pre- to post-test ($\beta = 2.17, SE = 0.49, p < .001$), regardless of condition. Children's objective AI literacy further increased from mid- to retention-test ($\beta = 2.26, SE = 0.50, p < .001$). Post hoc comparisons indicated that this effect was driven by an increase in the building condition ($\beta = 2.20, SE = 0.47, p < .001$). However, this effect may be due to an experimental error: children in the building condition completed their retention test without supervision, which may explain the unexpected increase relative to mid-test. We thus refrain from interpreting condition differences and treat the results as evidence of 7-week learning retention across both conditions. Pre-registered exploratory analyses of the three subscales are available in **Supp. F**.

4.1.3 Intrinsic motivation. There was a significant interaction effect between condition and phase ($\beta = -0.05, SE = 0.02, p = .04$) on children's intrinsic motivation (**Figure 3C**). Post-hoc comparisons indicated that intrinsic motivation was significantly higher in the building group than in the interaction group during building (phase 1; $\beta = 0.35, SE = 0.16$) but not during later interaction (phase 2; $\beta = 0.16, SE = 0.17, p = .33$).

This difference was primarily driven by the *interest/enjoyment* and *value/usefulness* subscales. For *value/usefulness* (**Figure 3D**), follow-up comparisons indicated that students in the building condition rated phase 1 ($\beta = 0.92, SE = 0.23, p < .001$), but not phase 2 ($\beta = 0.41, SE = 0.23, p = .09$), as significantly more useful than students in the interaction condition. For *interest/enjoyment* (**Figure 3E**), students in the building condition reported higher ratings across both phases ($\beta = 0.22, SE = 0.10, p = .01$), while enjoyment declined from phase 1 to phase 2 across both conditions (main effect of phase: $\beta = -0.12,$

$SE = 0.04, p = .002$). Perceived competence also decreased from Phase 1 to Phase 2 across both groups (main effect of phase: $\beta = -0.17, SE = 0.03, p < .001$), with no significant condition ($\beta = -0.03, SE = 0.08, p = .74$) or interaction effects ($\beta = -0.05, SE = 0.03, p = .09$). We did not find any significant effects regarding *information sufficiency* (p 's $> .53$) nor *pressure* (p 's $> .24$).

4.2 Perception (RQ3 & RQ4)

Across most social perception dimensions (i.e., competence, warmth, attachment, intention to use), likelihood ratio tests showed no evidence that condition, timepoint, or their interaction predicted students' perceptions of the Marty robot (all p 's $> .06$). Additionally, preregistered exploratory analyses showed no association of students' prior objective AI literacy with all perception outcomes (all p 's $> .30$). Full test statistics are available in **Supp. G**.

Significant effects centered on initial exposure dynamics. Across both conditions, perceived anthropomorphism decreased between pre- and mid-test ($\chi^2 = 19.33, p < .001$; $\beta = -1.68, SE = 0.43, p < .001$), but not between mid-test and post-test ($\beta = 0.04, SE = 0.45, p = .93$; **Figure 4B**), irrespective of condition ($\chi^2 = 0.05, p = .83$) or their interaction ($\chi^2 = 1.55, p = .46$). Additionally, students reported a significant increase in perceived safety between pre-test and mid-test ($\beta = 0.66, SE = 0.26, p = .02$), but not between mid-test and post-test ($\beta = -0.08, SE = 0.25, p = .76$; **Figure 4C**), and a significant decrease in discomfort between pre- and mid-test in the control condition ($\beta = -0.79, SE = 0.33, p = .03$), but not in the build condition ($\beta = 0.35, SE = 0.29, p = .23$; **Figure 4A**). However, because students in both conditions did not differ significantly in perceived discomfort at pre-test or mid-test, this interaction should be interpreted with caution. Preregistered exploratory analyses indicated that perceived anthropomorphism significantly explained variance in most perception measures (competence: $\chi^2 = 23.56, p < .001$; warmth: $\chi^2 = 5.02, p = .03$; safety: $\chi^2 = 7.91, p = .02$; discomfort: $\chi^2 = 5.66, p = .02$; attachment: $\chi^2 = 11.53, p < .001$; intention to use: $\chi^2 = 14.23, p < .001$), indicating that students who perceived the robot as more anthropomorphic regarded it as more competent ($\beta = 0.19, SE = 0.04, p < .001$), warmer ($\beta = 0.08, SE = 0.04, p = .02$), safer ($\beta = 0.19, SE = 0.04, p < .001$), more discomforting ($\beta = 0.08, SE = 0.03, p = .02$), and indicated greater attachment ($\beta = 0.07, SE = 0.02, p < .001$) and intention to use the robot ($\beta = 0.08, SE = 0.02, p < .001$), irrespective of condition or time point (**Supp. G, Figure S3**).

Lastly, preregistered exploratory analyses regarding students' intention to use Marty indicated that attachment and condition significantly improved model fit ($\chi^2 = 4.46, p = .04$):

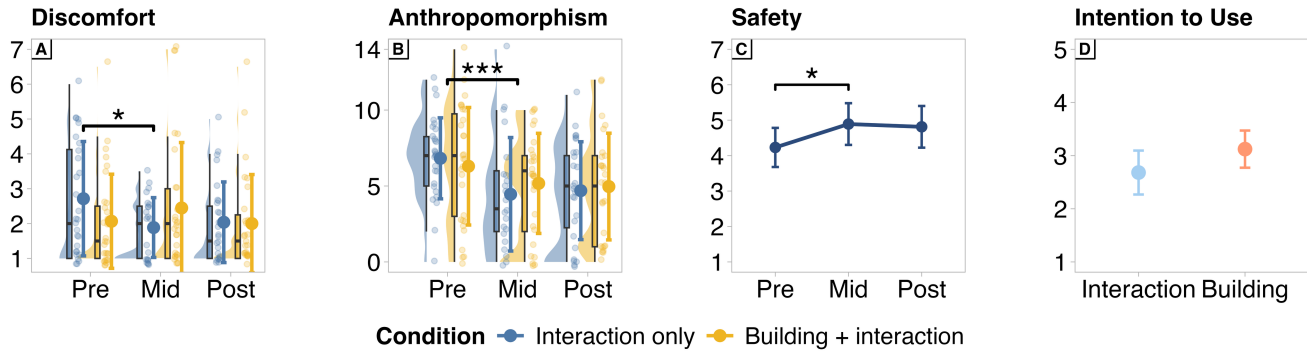


Figure 4. A) Significant interaction effect of time and condition on perceived discomfort. B) Significant decrease in anthropomorphism across time. C) Model-based estimated marginal means showing a significant increase in safety across time. D) Model-based estimated marginal means showing a significant main effect of condition on intention to use, when controlling for attachment.

attachment positively predicted students' intention to use the robot ($\beta = 0.60$, $SE = 0.07$, $p < .001$; **Supp. G, Figure S3**) and students in the building condition showed a higher intention to use the robot than students in the control condition ($\beta = 0.44$, $SE = 0.19$, $p = .02$; **Figure 4D**). However, since this effect also includes students' pre-test scores, it should be interpreted with caution.

4.3 Student and teacher classroom feedback (RQ5)

4.3.1 Student perceptions & engagement. Retrospective post-test ratings of overall project enjoyment, difficulty, and learning showed no significant differences between conditions (p 's $> .16$, **Supp. H**). However, open-ended qualitative feedback suggested distinct experiential patterns across phases. In Phase 1, students in the build condition framed Marty as a technical system, frequently highlighting technical progress, assembly, and early programming exploration. Conversely, students in the control group focused on immediate gameplay (e.g., penalty shootouts, obstacle courses) and commented more on AI-related learning and communication. This indicates that early positive feedback was not only shaped by the different activities but also by which content became salient in each condition. Additionally, building activities seemed to foster collaborative problem-solving and peer support, mitigating early group friction compared to the control condition. In Phase 2, builders cited AI-related learning and communication more frequently, whereas control students remained focused on basic interaction. Lastly, negative feedback was rather balanced between conditions and reflected specific implementation challenges rather than a general dislike of either condition. Moreover, negative feedback was not directly associated with social or competency perceptions of the robot. For example, among builders, negative feedback centered on time constraints and assembly difficulties, with students expressing frustration at having to stop building rather than disliking the task.

4.3.2 Teacher implementation experience. Due to the small sample size, results reported here are descriptive observations and do not reflect significant differences. Teachers rated students' engagement similarly high during Phase 1, but these ratings dropped in Phase 2 for the interaction condition. A similar drop in Phase 2 was observed regarding relevant student questions. The build condition was rated as more difficult to guide in Phase 1, both in terms of content and student behavior. In Phase 2, both difficulty ratings were more similar across conditions. Students' learning was rated as moderately to highly supportive, and these

ratings were relatively similar between conditions and phases (**Supp. H, Figure S5**).

Teachers' qualitative open-ended feedback further highlighted a trade-off between initial setup effort and long-term engagement. Teachers reported higher initial difficulty guiding the build condition due to content density and technical support demands, occasionally feeling overwhelmed by student troubleshooting. In this context, teachers noted uneven group progress and reported feeling overwhelmed by new components to teach without sufficient prior technical familiarity. Teachers expressed further concern that being unable to provide their usual support during assembly could lead to frustration among students. This suggests that successful implementation requires not only student scaffolding but also sufficient teacher preparation to respond to arising difficulties. In contrast, negative teacher feedback in the interaction condition focused primarily on student engagement and group dynamics. Teachers observed that collaboration reduced in certain groups as unengaged students withdrew from activities. Regardless, teachers expressed enthusiasm for the overall program, highlighting its success in offering rich opportunities for creative technical experimentation, programming, and meaningful peer collaboration. Teachers noted that the primary value of the build condition lay in cultivating focused, peer-supported engagement rather than manual construction skills.

5 Discussion

In this seven-week, classroom-based HRI study, we investigated how building and interacting versus only interacting with a pre-built social robot affects adolescents' AI literacy, intrinsic motivation, and social perceptions. Our results indicate that both conditions produced durable AI literacy gains that persisted seven weeks post-intervention. However, students who built their robots reported significantly higher intrinsic motivation, evaluating the curriculum as more useful and enjoyable. Regarding HRI outcomes, we observed no evidence of an IKEA effect: assembly did not inflate ratings of competence, warmth, discomfort, safety, or attachment. Instead, perceived anthropomorphism drove social evaluations, while early robot exposure recalibrated perceptions toward increased safety.

5.1 Learning (RQ1 & RQ2)

5.1.1 Lasting AI literacy gains in both groups (RQ1). Contrary to our expectations, we found no evidence that building a robot alongside learning AI concepts yielded additional objective or subjective learning gains over robot interaction alone. That robot interaction can support children's learning across several different learning contexts is well established [13], and our results contribute more data to this evidence base. Our findings demonstrate that physical assembly provides comparable learning outcomes and thus support the viability of robot building as an alternative pedagogical approach within social robot-supported education.

A notable finding was the marked divergence between subjective and objective literacy. While students expressed high self-efficacy, objective knowledge remained low (~35% post-test accuracy). This discrepancy aligns with established literature demonstrating that subjective self-reports are poor proxies for objective AI knowledge [24, 70, 122], possibly exacerbated here by Dunning-Kruger effects [i.e., overestimating low abilities, 34, 45, 53], and the difficulty of adapting adult measures for youth. Future work using age-appropriate objective measures is needed to determine how robustly this conclusion holds across different samples and instruments.

Beyond condition comparisons, our results advance the literature on educational HRI in three ways. First, while most prior work focuses on short-term interventions with younger children or with adults [44, 65, 73], we demonstrate that social robots can support sustained cognitive gains in adolescents that endure for at least seven weeks. Second, we provide a formal statistical evaluation of robot-supported AI literacy literature often constrained by small sample sizes and descriptive reports [119, 125]. Third, our findings add to a limited evidence base on robot building as a pedagogical approach [48, 126], demonstrating that construction activities can successfully be integrated into AI curricula. Future work should examine how robot-building impacts other facets of classroom dynamics, student learning processes, and longer retention periods.

4.1.2 Higher intrinsic motivation in building condition (RQ2). Building robots yielded significantly higher enjoyment and perceived usefulness than interacting with pre-assembled ones. This finding is particularly noteworthy because the motivational advantage persisted despite builders facing physical assembly problems, delays, and tool constraints, whereas the interaction-only group engaged in highly gamified activities (e.g., soccer tournaments, escape games). Grounded in productive struggle theory [121, 130], physical construction may have provided structured opportunities for students to develop perseverance and frustration tolerance [15], resulting in a more self-fulfilling experience.

However, this affective advantage did not translate into greater cognitive learning gains. Although intrinsic motivation frequently supports learning [37, 90, 129], affective benefits in HRI do not consistently translate into cognitive gains [13]. Interpreted through Chi's [23] taxonomy of active, constructive, and interactive learning, the shared collaborative activities in both conditions may have already met the threshold required for cognitive learning, leaving limited room for physical assembly to manifest in measurable differences. Nevertheless, educators may

leverage the heightened engagement of assembly to scaffold students through more abstract or challenging technical content, such as machine learning concepts or complex programming tasks.

5.2 Perceptions (RQ3 & RQ4)

5.2.1 Re-evaluating the IKEA effect in adolescent HRI (RQ3 & RQ4). Contrary to previous HRI findings, where assembly consistently produced more favorable evaluations [40, 43, 108, 126], we did not find evidence for increased socialness perceptions or attachment through assembling Marty. We propose that this discrepancy is not a failure of the DIY model, but rather a reflection of the specific intersection of robot morphology, task/context, and user development.

5.2.1.1 Robot morphology. Prior research indicates that the appearance of robots significantly affects social expectations [64, 81]. Unlike LEGO-based platforms [40, 114, 126], Marty's humanoid appearance may have induced a ceiling effect on social perceptions. Additionally, physical assembly without software "building", e.g., training a robot's AI algorithm [119] or building the underlying program [108], might have only provided half a building experience. Adding such components to physical building may offer valuable pathways to disentangle the role of hardware versus software building in the IKEA effect in HRI.

5.2.1.2 Children's development. While the IKEA effect is documented in adults and younger children [40, 43, 108, 114, 126], robot perception significantly changes across age [60, 68, 79], and ownership-through-effort mechanisms operate differently in children compared to adults [71]. For our 12–16-year-old participants, the lack of an IKEA effect may reflect a developmental shift toward utilitarian, task-oriented evaluations of technology. Consequently, assembly may have been processed as a routine technical classroom task rather than an identity-relevant act of creation that triggers an emotional bond.

5.2.1.3 The comparison/task. Lastly, unlike prior work that compared building to another building task [40], used software building/programming rather than hardware-building [108], or used pre-/post-test designs with no comparison group [43, 114, 126], we compared robot-building to an active, gamified robot-operation task. The high level of engagement in the control group's interactive tasks may have provided a similar sense of accomplishment to the building task, thereby neutralizing the IKEA effect of assembly.

5.2.2 Discomfort trajectories and intention to use. We cannot confidently interpret higher intention to use a robot when building it nor reduced discomfort when just interacting with it. Because between-group differences remained non-significant at all time points (discomfort), and the intention to use effect included pre-test scores, these effects may be at best small. To confirm whether intention to use and discomfort are driven by building experiences, future studies that isolate such effects in a larger sample are needed.

5.2.3 First encounters with a robot can alter perceived anthropomorphism and safety. Perceived anthropomorphism declined across the first four weeks in both conditions, possibly reflecting a shift in students' technical understanding of the robots' capabilities as students gained (technical) familiarity, in line with related work [14, 99, 104, 105, 116]. This reduction coincided with a significant increase in perceived safety, with

both variables stabilizing after Week 4. These findings suggest that initial encounters can drive perceptual recalibration and demonstrate that revealing a robot's mechanical nature can enhance perceived safety without diminishing social evaluations. However, different perceptual dimensions stabilize within different time frames [84], and future work should assess how students' perceptions of robots' socialness change across multiple interactions and the time frame over which perceptions may stabilize.

5.2.4 Anthropomorphism (and attachment) as predictors of perception. Extending previous work on different age groups [11, 46, 81, 103, 115], our findings confirm that perceived anthropomorphism remains a stable predictor of adolescents' socialness perceptions of robots. Furthermore, attachment positively predicted intention to use. However, our findings also suggest that anthropomorphism increased perceived discomfort, suggesting that uncanny valley effects [47, 69] may be present even with low-level humanoid features during extended interactions.

5.3 Broader implications for educational social robotics

Our findings validate the practical and economic viability of low-cost DIY robotics in formal education. Demonstrating that hands-on assembly can match pre-assembled robots in cognitive gains while exceeding them in student motivation provides a compelling rationale for adopting open, affordable toolkits over expensive, proprietary systems (e.g., NAO, Pepper).

Conceptually, these results advocate for repositioning social robots from rigid tutors or pre-packaged tools to more diverse and participatory roles [93]. Involving students directly in construction can foster technical transparency, trust, and user agency, key prerequisites for stakeholder acceptance and sustainable classroom integration [10, 38, 59, 101, 102, 123]. Emerging platforms designed for physical customization [Blossom, 107], movement puppeteering [YOLO, 9], or community app development [Reachy mini, 92] demonstrate that the technical infrastructure for participatory robotics exists; future research must now evaluate how to implement and scale these agents across diverse educational contexts.

5.4 Limitations

In-the-wild longitudinal HRI studies present expected and unexpected challenges [127], which must be considered when interpreting findings. Technically, an LLM failure during Week 7 required replacing Marty's verbal interaction with additional programming activities for the interaction condition. This introduced experiential variance between conditions, visually reflected in higher programming knowledge among control students. Furthermore, natural variations in classroom pacing across sessions introduced noise into the curriculum implementation. Finally, Marty provided an accessible platform, but its proprietary nature limits open-source customization and community adaptation compared to platforms like Blossom [107].

Pedagogically, our deployment structure across adjacent classrooms introduced teacher-level confounders. Because both teachers delivered the build condition first, their increased familiarity with the curriculum may have unintentionally elevated instructional quality for the subsequent interaction condition. Additionally, enforcing experimental consistency between rounds

prevented teachers from making adjustments to resolve timing or difficulty issues during the second phase, causing teacher frustration during control sessions. While future experimental designs should counterbalance condition order, real-world implementation depends on teacher-led adaptation. Future research should observe natural, unconstrained classroom deployments over extended periods to investigate how teachers adapt these toolkits in practice and how those modifications influence student engagement.

Measurement limitations also dictate a cautious interpretation of cognitive learning gains. Although objective literacy improved significantly, overall post-test scores remained low (~35% correct). This stems from a broader field challenge: the lack of validated, age-appropriate objective AI literacy measures independent of specific curricula. Although we adapted adult instruments to reduce difficulty, teacher feedback confirmed that several items remained overly complex, meaning scores likely underestimate students' true knowledge gains. Future work should develop validated, age-appropriate objective measures of AI literacy, adapted to different stages of students' digital education, which varies across countries [96]. Additionally, although we included attention checks and emphasized careful completion, fluctuating survey completion times across measurement points suggest that student response effort varied over the seven weeks. Future studies should explore alternative strategies to protect against careless responding, such as linking assessments to in-class grades.

Finally, in line with Simons and colleagues [100], our sample characteristics limit the generalizability of our findings. Participants were drawn from a single German comprehensive school with relatively high socioeconomic status and a male-skewed gender balance, a key factor given persistent gender disparities in Science and Technology engagement [42, 72]. Caution is therefore required when extrapolating these results to lower-SES populations, gender-balanced classrooms, different cultural settings, or alternate robotic platforms [13, 26]. We are currently extending this research line using the Blossom robot in a remote Australian context with younger children to explore the extent to which these findings hold with a different robotic platform in a different sociocultural and geographic context.

5.5 Conclusion

This study compared interactions with a pre-built versus self-built robot across a seven-week AI literacy curriculum. Both approaches supported learning, but students experienced significantly greater intrinsic motivation, perceived usefulness, and enjoyment when they assembled the robots themselves. Our results align with prior work demonstrating the value of long-term, in-the-wild robot deployments for learning, and extend most such deployments by providing evidence for objective and subjective learning gains in adolescence [28, 47, 55, 73]. Joining Marsh and colleagues [71], our findings challenge the IKEA effect in adolescent HRI. Furthermore, we demonstrate that initial robot exposure calibrates students' perceptions. These results highlight the theoretical and practical value of low-cost DIY toolkits, repositioning social robots not merely as tools for interaction, but as agents to construct.

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